

ARMIN ESHAGHI

EMBEDDED C/C++ | COMPUTER VISION | SOFTWARE DEV

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PROFESSIONAL SUMMARY

Algorithm engineer with 5+ years of experience. Building performance-sensitive C/C++ and Python systems for automotive ASICs and biomedical robotics. Developing reusable embedded C libraries, validating ML inference on iND881 (custom silicon), and evaluating real-time performance and accuracy across hardware/software boundaries. University of Toronto Masters (MAsc) in Robotics and IEEE-published researcher.

CORE TECHNICAL SKILLS

Programming: C, C++, Python, Linux, Git, Docker, NumPy, OpenCV

ASIC / Embedded: iND881 ASIC, embedded systems, hardware/software integration, real-time inference, performance and accuracy analysis, validation; Cadence P6 DSP (in progress)

ML Optimization: MobileNet (ASIC testing), quantization, pruning, PyTorch, TensorFlow

Engineering: Reusable C libraries, geometric transformations, low-latency data capture, MISRA-C, mentoring

PROFESSIONAL EXPERIENCE

Senior Perception Algorithm Engineer | indie Semiconductor Inc.

Mar 2022 - Present

- Own and maintain eWARP®, the company's core geometric-transformation engine, and build reusable C libraries adopted across product lines and engineering teams.
- Develop image-correction and computer-vision algorithms in Python and C/C++ for automotive edge processors; evaluate real-time image quality and sensor performance, supporting client acquisition.
- Tested MobileNet models on custom ASIC to validate accuracy and performance of on-chip inference; currently extending deployment workflows to Cadence P6 DSP and evaluating YOLOv5.
- Mentor intern algorithm developers through technical guidance, code reviews, and MISRA-aligned practices, reinforcing production-quality embedded software.

Software Engineer Consultant | Toronto Rehab Institute, UHN

Dec 2022 - Jun 2023

- Built an OpenCV gait-analysis and fall-prediction pipeline that generated quantitative data for fall-prevention.
- Integrated C++, Python, cameras, motion sensors, and hardware triggers into a synchronized real-time data-capture system; developed Linux workflows that reduced post-processing effort for large datasets.

Research Assistant | University of Toronto

Sep 2018 - Dec 2021

- Built an automated single-cell surgery platform combining real-time 3D segmentation, coordinate extraction, and robotic control, improving performance 110% over manual operation and producing publications.
- Designed and experimentally validated a C++ visual-servo controller that achieved sub-micron accuracy for automated micro-injection on live biological samples.

EDUCATION & CREDENTIALS

MAsc, Mechanical Engineering (Robotics) | University of Toronto

2021

Thesis: Automation of Real-Time 3D Image-Based Visual Servo Control for Single Cell Surgery

BASc, Mechanical Engineering | University of Toronto

2018

Minors: Robotics and Biomedical Engineering

Publication: "Automated Real Time Image Based Visual Servo Control of Single Cell Surgery," IEEE ICMA, 2020

Training: Harvard TinyML (TensorFlow Lite, edge deployment), 2022 | fast.ai Practical Deep Learning, 2023